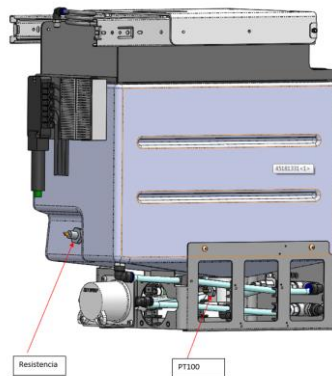


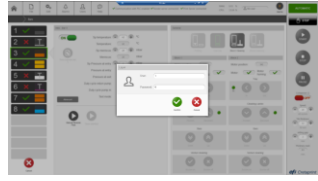
USER MANUAL: INK HEATER TANK SYSTEM

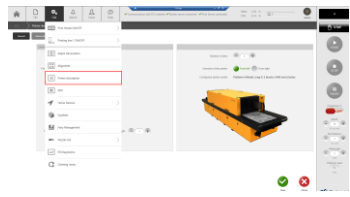
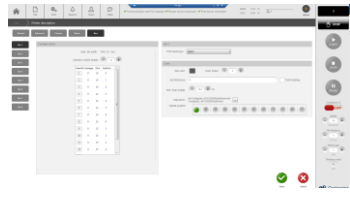
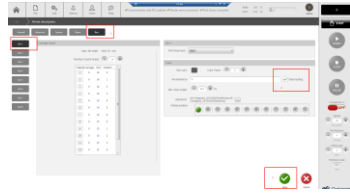
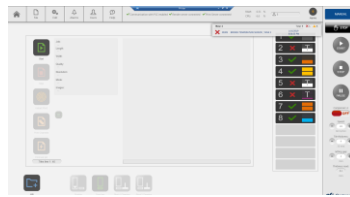
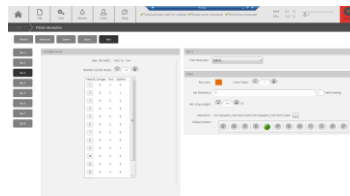
Ink Heater Tank management is a feature to heat ink while print bar is recirculating. This system is based on resistor and temperature sensor located at heater tank installed as is shown on next picture.



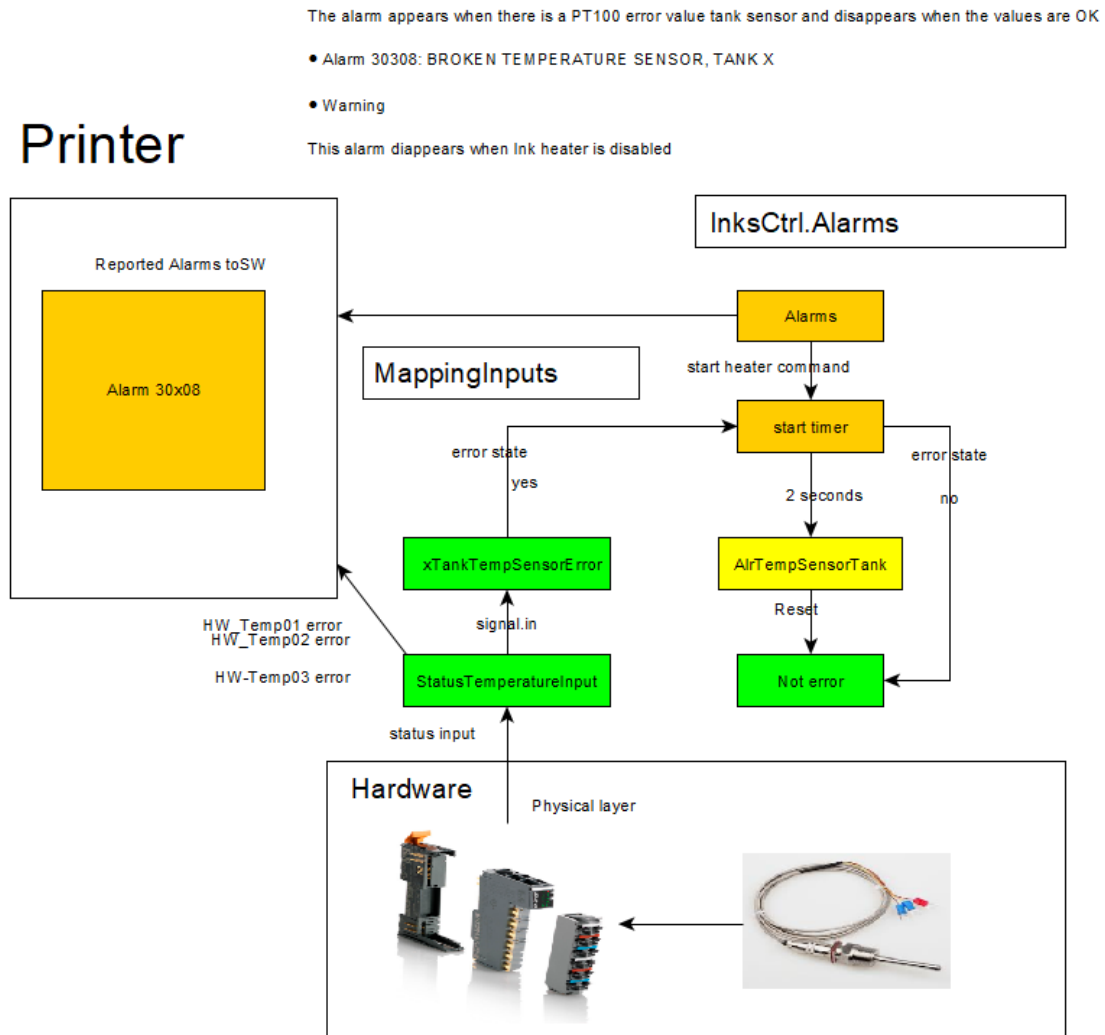
Configuration: ink heater tank (activation / deactivation)

To activate the ink tank heater, the user must follow these steps:

Process	Description	
Main window	Stop inks recirculation on print bar that will be enabled for tank heating	
	User access permissions: Test mode needs to be disabled with high user access	
	Disable Test mode, print bar in test mode needs to be disabled before tank heating starts	

Open Printer Description and click on Bars menu		
	In bars menu window, remark checkbox for heating ink installed in tank	
	When there are no alarms stopping recirculation on printbar, then heater tank starts warming up.	
Alarms	Heating process has sensor checker and status input temperature is reporting this alarm just in case of open circuit on PT100 installed.	
Alarm 30308: BROKEN TEMPERATURE SENSOR, TANK 3 (x).	It appears when there is a PT100 error value of tank sensor and disappears when the values are properly	
	Alarm disappears when ink tank heater is disabled. It's a warning alarm and printer doesn't stop with it.	

A Specific functional chart is explaining alarm about ink tank heater



X20AT4222 module is equipped with 4 inputs for PT100/PT1000 resistance temperature measurement and plate X2X where are connected 4 Resistor Temperature Inputs PT100 to PLC and, in PLC is implemented a functional block to detect open circuit status

Mapping physical temperature inputs

```

gBlock[1].In.xModuleTemp01Ok      := X20AT4222_114.ModuleOk;
gInksInputs[6].iTemperatureRaw    := X20AT4222_114.TempInput[1];
gInksInputs[6].xTempSensorError   := StatusTemperatureInput(X20AT4222_114.Status01, 1);
gInksInputs[5].iTemperatureRaw    := X20AT4222_114.TempInput[2];
gInksInputs[5].xTempSensorError   := StatusTemperatureInput(X20AT4222_114.Status01, 2);
gInksInputs[4].iTemperatureRaw    := X20AT4222_114.TempInput[3];
gInksInputs[4].xTempSensorError   := StatusTemperatureInput(X20AT4222_114.Status01, 3);
gInksInputs[3].iTemperatureRaw    := X20AT4222_114.TempInput[4];
gInksInputs[3].xTempSensorError   := StatusTemperatureInput(X20AT4222_114.Status01, 4);
    
```

The Implementation code [StatusTemperatureInput: USINT, USINT] is just detecting open circuits states on X20AT4222 modules. It means that PT100 sensor is not detected by PLC

Sometimes, cannot detect PT100 connected with resistance in work range correct but incoherence temperature. With same conditions all printbars have to measure same values aprox, cannot detect: 27 degrees in printbar 1st and 10 degrees in the rest.

```
FUNCTION StatusTemperatureInput

    //If Module is NOT OK, directly It returns an alarm, Block.Al.xModuleTemp0X

    //Channels iInputTemperatureNumber from 1 to 4 in iStatusInputTemp01
    iStatusTemperatureInput:= SHR(iStatusTempInput01, (iTemperatureInputNumber-1)*2) AND 16#03;
    // Output boolean, value 3 means that there is an opencircuit in PT100
    IF iStatusTemperatureInput > 2 THEN
        StatusTemperatureInput:= TRUE;
    ELSE
        StatusTemperatureInput:= FALSE;
    END_IF

END_FUNCTION
```

Alarm implementation for heating sensor

```
//Alarm temperature gauge broken in heater or temperature gauge broken HEATER TANK. Error detected by the plc.
//HEATER TANK
TON_DelayTempTankSensor(IN:= data.status.xBarInstalled AND
                           data.signals.in.xTankTempSensorError AND
                           data.status.xEnabledTankHeater,
PT:= T#2S);

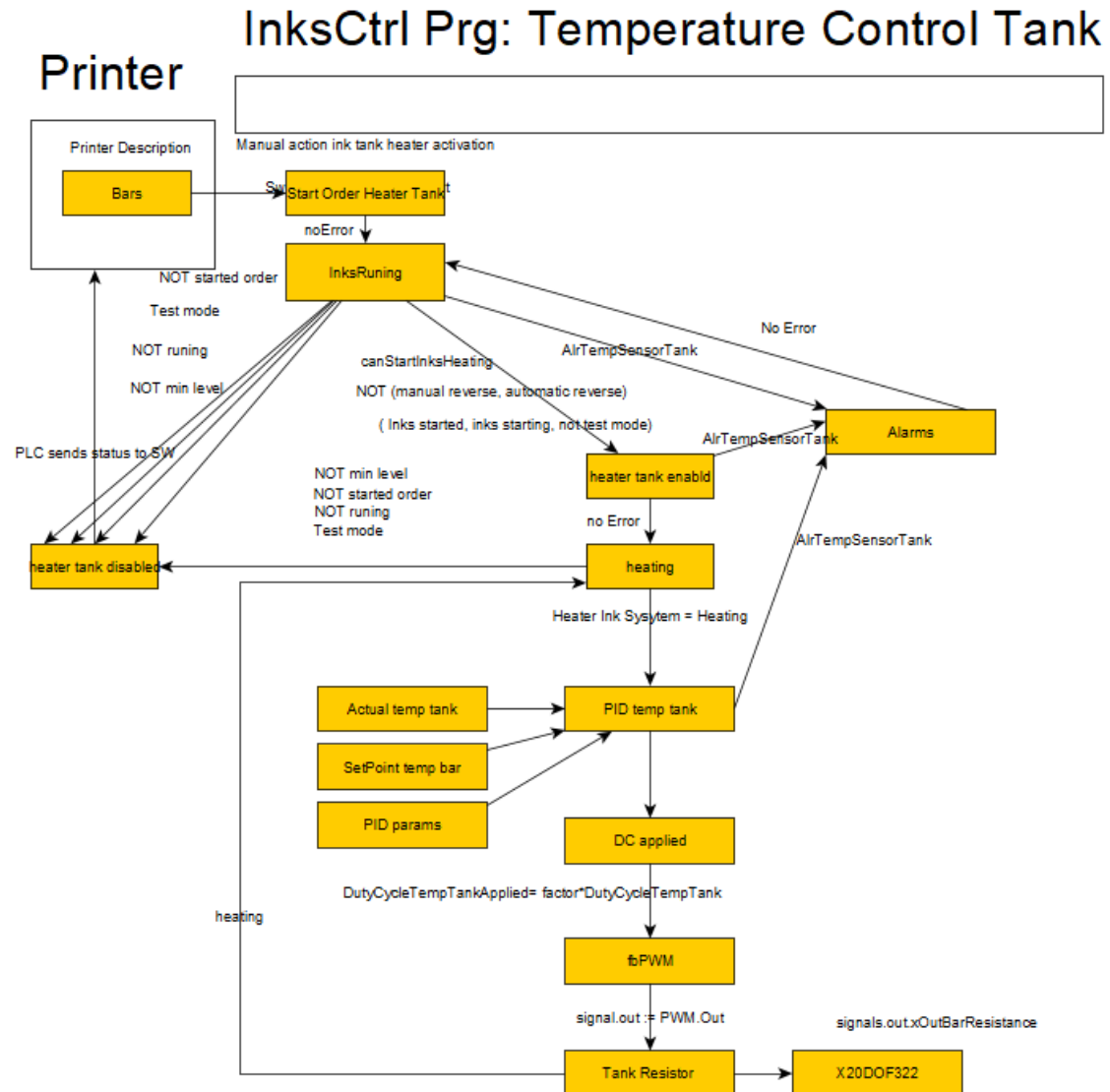
// If not installed bar en It is not an alarm
IF TON_DelayTempTankSensor.Q THEN
    data.Al.xAlrTempSensorTank := TRUE;
ELSIF data.commands.xErrorRst OR (NOT data.status.xBarInstalled) OR (NOT data.status.xEnabledTankHeater) THEN
    data.Al.xAlrTempSensorTank := FALSE;
END_IF
```

It is based on a timer that is activated by the result of our functional block explained before.

Latest step, PLC is reporting alarms to SW by comms [PLC2Kernel]

```
3 | HoldingRegister[2011].0:=gInks[7].Al.xAlrTemp;
  | HoldingRegister[2011].1:=gInks[7].Al.xAlrTempSensorTank;
  | HoldingRegister[2011].2:=gInks[7].Al.xAlrTempSensorTank;
7 | HoldingRegister[2011].4:=gInks[7].Al.xAlrVacuumSensor2;
```

A Specific functional chart is explaining how to control temperature on ink tank heater



When start tank heater order comes from SW to PLC then ink system starts to heat ink tank and just in case alarms, it stops.

PID is configured to calculate a Duty Cycle and a PWM functional block is connecting resistors with X20DOF322 that is generating a signal for these relays



Conclusions

- We need implement on GUI any indicator for heater tank monitoring, temp tank current

Process	Description	
Main window	Heater tank state shows when it is working properly Kernel has Duty Cycle per bar to get info about it [(DC!=0)THEN ON ELSE OFF] Edu purposes, Heater tank never reaches bar temp, for this reason temp tank on gui allows to users also identify when heater tank is working properly Pending to evaluate: Can it be noise for customers???	